Project Documentation: Smart City & Urban Sustainability Analytics

Goal: Sustainable Development Goal 11 (SDG 11) – Sustainable Cities and Communities

Location: Bengaluru, India

Scope: 150,000+ data records across Transport, Housing, Environment, and Infrastructure.

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1. Project Overview

This project leverages data analytics and machine learning to evaluate the urban sustainability of Bengaluru. By integrating four distinct urban sectors, the project provides a composite **Sustainability Index** to identify high-risk zones and key drivers of city health.

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2. Analytical Approach

A. Data Preprocessing & Cleaning

- **Integration:** Merged real-time Environmental (AQI) data with historical Traffic, Housing, and Waste/Water datasets using spatial and temporal keys (Station & Hour).
- **Scaling:** Handled over 10 million raw potential combinations by downsampling to a stable **150,000 rows** for efficient processing.
- **Imputation:** Used **Median Imputation** to handle missing values in PM2.5 and Traffic Volume, ensuring a 100% clean dataset for modeling.

B. Feature Engineering

- **Sustainability Index (Target Variable):** Created a weighted score (30% Air Quality, 30% Transport, 40% Waste Management) to classify areas as **High, Medium, or Low** sustainability.
- **Risk Score:** Engineered a new metric combining high pollution levels and high traffic volume to flag critical urban hotspots.

C. Segmentation & Clustering

- **K-Means Clustering:** Segmented the city into three distinct "Urban Zones" (Cluster 0, 1, and 2) based on infrastructure efficiency and environmental impact.

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3. Key Performance Indicators (KPIs)

The project tracks the following metrics as indicators of SDG 11 progress:

1. **Environment:** PM2.5 Concentration (μg/m³).
2. **Transport:** Average Traffic Speed & Public Transport Usage (%).
3. **Infrastructure:** Waste Management & Water/Sanitation Efficiency Scores.
4. **Economic:** Average Housing Price per locality.

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4. Model Results & Insights

Machine Learning Performance

- **Algorithm:** Random Forest Classifier.
- **Accuracy: 99.99%**.
- **Insight:** The model identified **PM2.5 levels** and **Average Traffic Speed** as the two most significant predictors of urban sustainability.

Key Findings

- **Traffic-Pollution Link:** EDA confirms a direct correlation between traffic volume peaks and hazardous air quality spikes.
- **Regional Variance:** High-density hubs (like Silk Board) show higher Risk Scores compared to developing residential zones (like Hebbal).

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5. Project Key Takeaways

1. The Primary Driver

Our analysis identifies **PM2.5 (Air Quality)** as the most critical factor in urban sustainability. This suggests that to meet **SDG 11 Target 11.6**, Bengaluru must prioritize air pollution sensors and "Low Emission Zones" in high-risk clusters like **Silk Board**.

2. The Traffic Threshold

The **Segmentation (K-Means)** plot revealed that once traffic exceeds **40,000 units** (Cluster 2), the urban "load" becomes critical. Policy interventions should focus on expanding **Public Transport Usage** in these specific high-volume hubs to maintain sustainability.

3. Predictive Planning

With a **99.99% accurate model**, city planners can now "test" future scenarios. For example, if a new housing project increases traffic by 10%, the model can instantly predict if that ward's sustainability level will drop from "High" to "Medium."

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6. Future Recommendations

- **Smart Traffic Management:** Implement AI-driven signals in "Low Sustainability" zones to improve average speed and reduce idling emissions.
- **Green Infrastructure:** Prioritize waste processing units in high-density wards to balance the Urban Risk Score.

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7. Technical Stack

- **Language:** Python (Google Colab).
- **Libraries:** Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib.
- **Visualization:** Power BI Desktop for the final interactive dashboard.